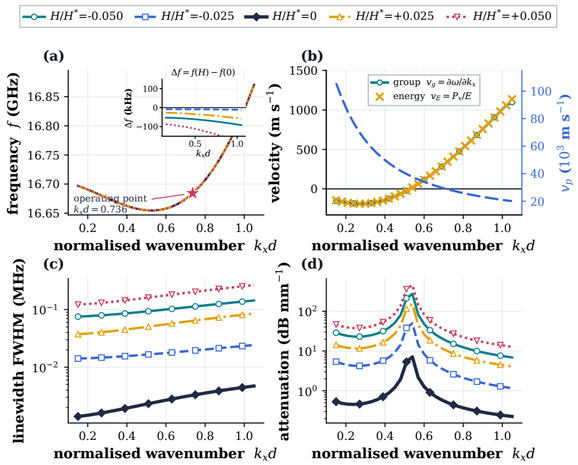
Paper A -- figure atlas (1/2)

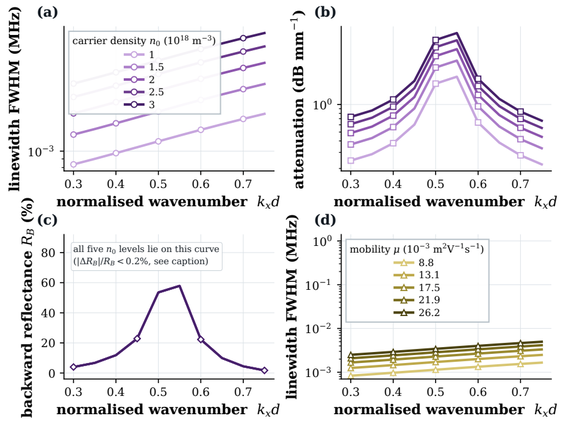
Figure 1 Dispersion and wave speeds



The five H/H* levels superimpose over the whole zone; the inset resolves the residual splitting (<60 kHz on a 16.7 GHz carrier). (b) uses the full wavevector for v_p, a numerical d(omega)/dk_x for v_g and an integrated Poynting/energy ratio for v_E; the three change sign near k_x d = 0.55, the backward-branch signature. Attenuation is 2|Im k_x|, never zero.

source: fig01_dispersion.csv

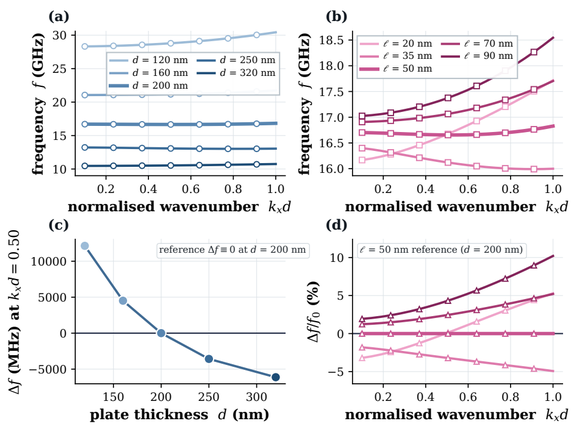
Figure 2 Carrier density and mobility



n_0 scales the linewidth and the attenuation but not the backward reflectance: (c) collapses five densities onto one R_B curve with a maximum of 58 % near k_x d = 0.55. Only certified rows are drawn, so the curves stop where the backward port collapses.

source: fig02_material.csv

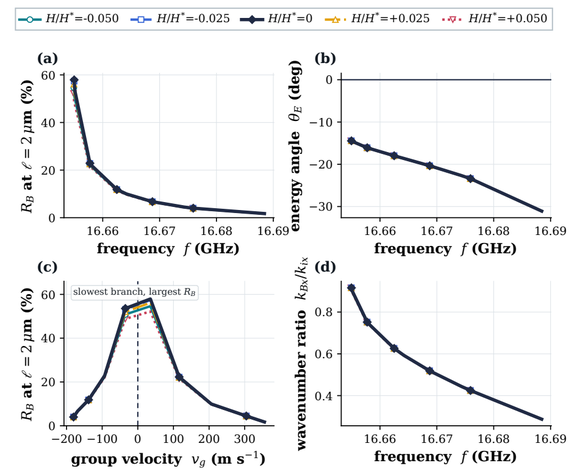
Figure 3 Thickness versus gradient length



Thickness sets the branch (17 GHz between 120 and 320 nm) while the gradient length perturbs it by at most 10 % and only at large k_x d. The two families are separated, as they must be, because they scale differently.

source: fig03_geometry.csv

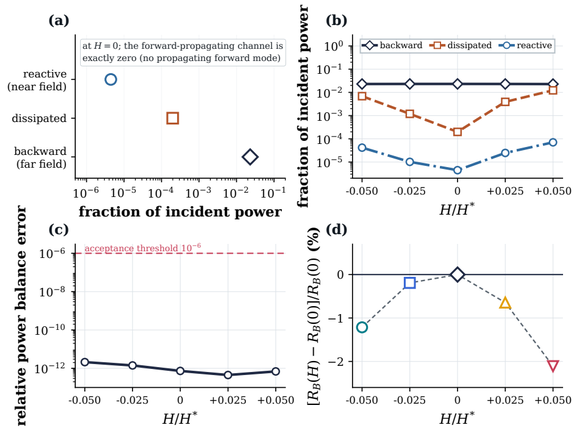
Figure 4 Backward conversion on the guided branch



All 50 branch-tracking rows are certified (0<=R_B<=1, |theta_E|<80 deg, balance<1e-6, edge residual<1e-8). Conversion peaks where v_g -> 0, so the effect is a slow-mode effect. Rows at |theta_E| -> 90 deg are not reflectances and are absent by construction.

source: fig04_frequency.csv

Figure 5 Energy budget of the edge problem

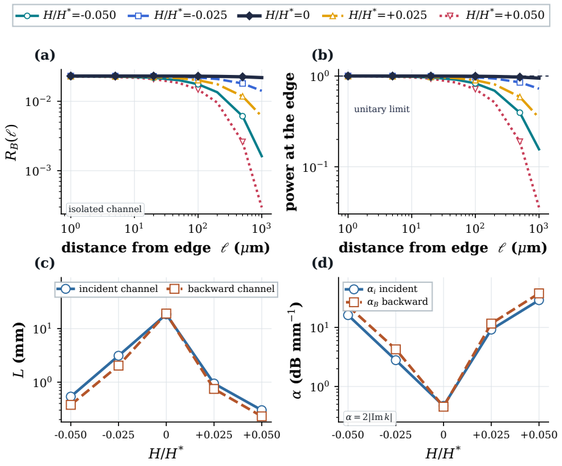


Incident power is normalised to one and split into a backward far field, dissipation and a reactive near field. The forward-propagating channel is exactly zero and is stated as such. The balance error stays at 1e-12 and the coupling moves R_B by at most 2.5 %.

source: fig05_channel.csv

Paper A -- figure atlas (2/2)

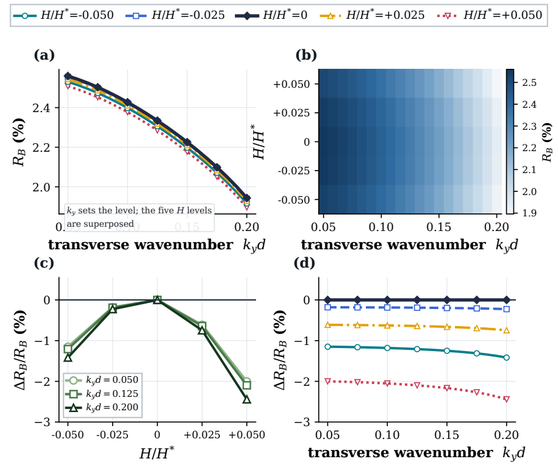
Figure 6 Distance, survival and attenuation



The backward coefficient decays with the observation distance and the survival factor never exceeds unity. Lengths and loss coefficients in (c)-(d) come from the solved complex spatial roots, alpha = 2|Im k|, and are positive everywhere.

source: fig06_attenuation.csv

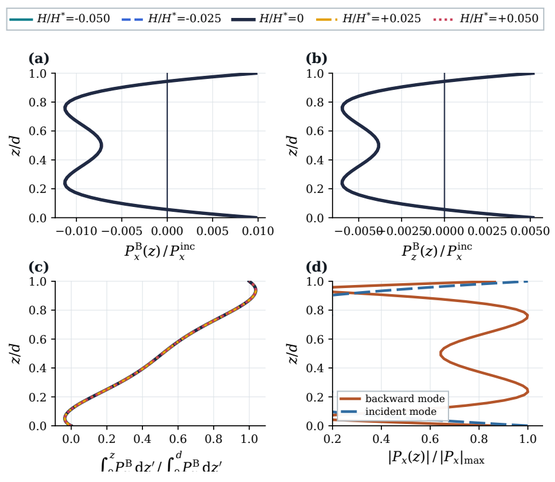
Figure 7 Coupling sensitivity in (k_y, H)



The transverse wavenumber sets the level and the coupling shifts it by at most 2.5 %. Panel (b) draws the solved grid cell by cell: nothing outside the computational box is interpolated.

source: fig07_coupling.csv

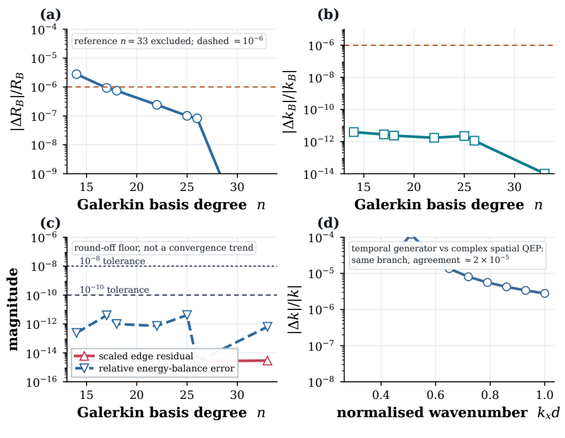
Figure 8 Through-thickness flux distribution



The x-component of the backward flux changes sign inside the plate while its thickness integral stays negative, and the z-component does not vanish at the free surfaces. That is why the reactive near field has to be carried in the budget rather than dropped.

source: fig08_boundary.csv

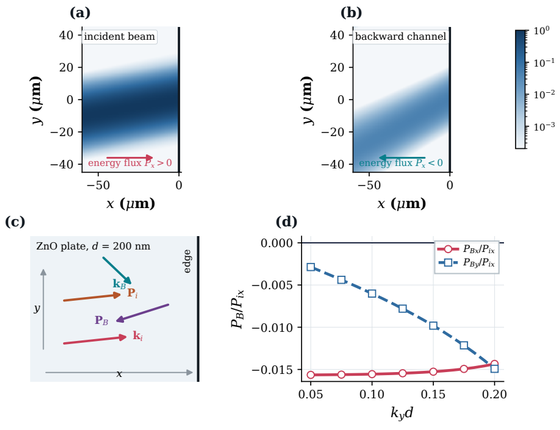
Figure 9 Convergence and cross-checks



Reflectance and backward wavenumber converge to the round-off floor by n = 26; residuals stay below 1e-8 and 1e-10. Panel (d) compares two solution routes for the same dispersion relation -- not two independent models -- and they agree to 2e-5.

source: fig09_validation.csv

Figure 10 Solved field and flux directions



Incident and backward intensity share one logarithmic scale. Every arrow in (c) is taken from the solved thickness-integrated flux: incidence at +10 deg, backward flux at -27.9 deg, both components negative.

source: fig10_field_maps.csv